

B.2.6 A+ 220-1201 Core1 Exam Practice

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Time Spent: 50:57

Score: 80%

Passing Score: 80%



Question 1

✓ Correct

A systems engineer is building a virtual machine to run a service with an 8 core CPU and 500 GB of random access memory to function properly. Where would these two hardware requirements be listed?

- ☐ Legacy software/OS
- ☐ Test development
- ☒ Resource requirements ✓ Correct
- ☐ Security requirements

Explanation

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Security requirements are incorrect because the amount of central processing units (CPUs) and random access memory (RAM) a system needs will not impact the security of a system.

Legacy software/OS is incorrect because a list of legacy software and OSs would not encompass hardware like CPU and RAM.

Test development often uses virtual desktop platforms based on some guest OS hypervisor.

Related Content

resources\questions\pt_core1_220_1201_4_1_013.question.xml

Question 2

✔ Correct

A user reports that their reporting resource is not working. They mentioned an error message that they did not write down but are certain that a "HTTP request failed" was a part of the error. What specialized support team will the user contact to resolve this issue?

- ☐ File server team
- ☒ Web server team ✔ Correct
- ☐ Print server team
- ☐ Mail server team

Explanation

A web server provides client access using HTTP or its secure version (HTTPS). Given the limited information, this would be the first team to reach out to.

Mail servers primarily use Simple Mail Transfer Protocol (SMTP), which specifies how the user delivers an email from one mail domain to another domain.

Print servers rely heavily on Server Message Block (SMB) protocol for file and printer sharing on a Windows network.

File servers also rely heavily on Server Message Block (SMB) protocol for file and printer sharing on a Windows network.

Related Content

7.1.3 Web Servers

resources\questions\pt_core1_220_1201_2_3_002.question.xml

Question 3

✔ Correct

A user's phone charge only lasts two hours. The phone is new and has not suffered any damage. What should the user check next?

- ☐ Charging cable
- ☒ Battery utilization ✓ Correct
- ☐ Digitizer
- ☐ Overheating

Explanation

If a battery is new and the phone has not suffered damage, the next step is to check battery utilization. Battery utilization will indicate if a particular app is draining the battery excessively.

Mobile devices use passive cooling and can be quite warm to the touch when starting to overheat. There is no indication of a physically hot device in the scenario.

Symptoms, such as the touch screen not responding to input, indicate a problem with the digitizer.

If the mobile device were not fully charging, then the user would need to examine or perhaps replace the charging cable. However, there is no indication that the phone is not charging; therefore, the problem is that the charge is not lasting.

Related Content

 9.4.1 Power and Battery Issues

 9.4.4 Screen and Calibration Issues

resources\questions\pt_core1_220_1201_5_4_004.question.xml

Question 4

✔ Correct

A user has a newly issued smartphone that has corporate communication software installed. Which of the following options will NOT sync with Microsoft Exchange?

- ☐ Contacts
- ☐ Email
- ☐ Calendar

☒ Texts ✔ Correct

Explanation

Texts are messages that are sent via cellular service. While there are sync methods available, they are not synced with Exchange.

Most email systems store messages on the server, and the client device is used to manage them. A client device that uses technology such as ActiveSync can store a local copy.

A contact is a record with fields for name, address, email address(es), phone numbers, notes, and more. Contacts are synced with Exchange ActiveSync.

A calendar item is a record with fields for appointment or task information, such as subject, date, location, and participants. Calendars are synced with Exchange ActiveSync.

Related Content

 9.2.3 Types of Data to Synchronize

 9.2.4 Email Configuration Options

resources\questions\pt_core1_220_1201_1_3_008.question.xml

Question 5

✔ Correct

A technician needs to upgrade components in a laptop. Which component can the technician install without the use of tools?

- ☐ Wireless card
- ☐ SSD drive
- ☐ Memory

☒ Battery ✔ Correct

Explanation

Laptop computers use removable, rechargeable Lithium-ion (Li-ion) battery packs. Some batteries require the use of tools to access, while many batteries are easily removable with a spring lock mechanism on the back of the laptop.

A solid-state disk drive (SSD) is internal and would require tools to replace. Memory is under a removable panel or by removing the entire back cover.

Memory is internal and would require tools to access. Memory is under a removable panel, which can be accessed by removing the entire back cover or by removing the keyboard.

A wireless card is an internal add-on card accessible by a removable panel on the bottom of the laptop or by removing the entire back cover of a laptop.

Related Content

9.3.2 Battery Replacement

resources\questions\pt_core1_220_1201_1_1_001.question.xml

Question 6

✔ Correct

A startup legal office has specific software needs that are unique to the practice of law. The law firm partner would like to pay software developers to create a custom application but is concerned that the firm does not have the funds. The firm's IT manager has recommended that it use existing applications instead of paying for a new app to be developed. What service is the IT manager suggesting to the firm's partner?

- ☐ Platform as a service (PaaS)
- ☐ Virtual desktop infrastructure (VDI) in the cloud
- ☒ Software as a service (SaaS) ✓ Correct
- ☐ Infrastructure as a service (IaaS)

Explanation

Software as a service (SaaS) is correct. The law firm will pay for a certain number of software licenses to use already developed software to meet their unique needs.

Platform as a service (PaaS) is incorrect because the law firm does not need a service provider for infrastructure and software.

Infrastructure as a service (IaaS) is incorrect because the law firm has its infrastructure.

Virtual desktop infrastructure (VDI) in the cloud does not pertain to this question because the law firm does not need a means of provisioning corporate desktops using virtual machines.

Related Content

8.2.5 Common Cloud Service Models

resources\questions\pt_core1_220_1201_4_2_007.question.xml

Question 7

✔ Correct

The TIA/EIA-568-C Commercial Building Telecommunications Cabling Standards have specific recommendations for healthcare facilities with Power over Ethernet (PoE) 802.3bt installations, and for running distribution system cables to wireless access points. Which Cat does this refer to?

- ☐ Cat 5e
- ☐ Cat 5
- ☐ Cat 6
- ☒ Cat 6A ✓ Correct

Explanation

Cat 6A is recommended for healthcare facilities since it supports 10 Gbps over 100m, but the cable is bulkier and heavier than Cat 5e and Cat 6. The installation requirements are more stringent.

Cat 5 cable supports the older 100 Mbps fast Ethernet standard. It is no longer commercially available, so a technician will need to rewire the network cable with Cat 5 to support Gigabit Ethernet.

Cat 5e cable supports Gigabit Ethernet links for client computers, but can now opt to install Cat 6 cable.

Cat 6 cable is more reliable than Cat 5e and supports 10 Gbps but over a reduced range.

Related Content[5.3.4 Cat Standards](#)[resources\questions\pt_core1_220_1201_3_2_012.question.xml](#)

Question 8

✔ Correct

Users reported an incident where they could not connect to network shares on a specific file server but could utilize all other resources and the internet. After logging into the server locally to check, it seems to be up and connected to the network. There is no response after going to a user's station and using the ping command to contact the server by the documented IP address. What should be the issue to investigate next?

- ☒ Check the server's IP address. It may have a DHCP-provisioned address instead of a static address. ✔ Correct
- ☐ Ensure the server has a Class A IPv4 address.
- ☐ Check the user's default gateway to ensure it can communicate to remote networks.
- ☐ Check to see if the user workstation has been given an APIPA address.

Explanation

If the server locally appears online and connected, it has an address, but likely one provisioned from DHCP if workstations cannot ping the server by the documented IP address.

The user workstation would not likely reach all other resources if it had an APIPA address.

If the user station can connect to other resources, especially the internet, there is likely no issue with the gateway.

It may not be a Class A address, depending on the type of network structure. A technician can also use class B and C addresses for private addressing.

Related Content

resources\questions\pt_core1_220_1201_2_6_005.question.xml

Question 9

✔ Correct

Which of the following allows reducing the number of cables to the minimum required to minimize clutter within the chassis and improve airflow and cooling?

- ☐ Redundant power supply
- ☐ Thermal pad
- ☐ Heat sink
- ☒ Modular power supply ✔ Correct

Explanation

A modular power supply has power connector cables that are detachable from the unit.

A redundant power supply requires a compatible motherboard. This configuration is more common in server systems than on desktop PCs. Typically, each PSU plugs into a backplane on a server and is hot-swappable.

A heat sink is a block of copper aluminum with fins. The fins expose a larger surface area to the air around the component to achieve a cooling effect by convection.

A thermal pad is a solid compound at room temperature but softens when heated.

Related Content

3.1.6 Modular Power Supplies

resources\questions\pt_core1_220_1201_3_6_002.question.xml

Question 10

✔ Correct

While trying to set up for a presentation, the wall-mounted display will not mirror the laptop display. The technician traces the long HDMI cable to find that it runs on the floor, and it is evident that users have repeatedly run over it. What will likely fix the problem?

☒ Use a new HDMI cable. ✔ Correct

☐ Replace the PSU.

☐ Turn on the laptop.

☐ Plug the cable securely in.

Explanation

A cable that has physical damage will likely be the cause of the problems. Replacing the high-definition multimedia interface (HDMI) cable and finding another way to run the wires to avoid damage in the future will likely solve this problem.

A faulty power supply unit (PSU) on the laptop would not cause problems displaying on a second monitor.

The damaged HDMI cable will likely still not work, even if plugged in securely on both ends.

The laptop is already on since the display on the laptop is visible and not correctly mirroring on the display monitor.

Related Content

4.3.6 Troubleshoot Missing Video Issues

resources\questions\pt_core1_220_1201_5_3_002.question.xml

Question 11

✔ Correct

A user's computer is not connecting to Wi-Fi. After gathering more information, the technician finds out that the user just returned from a work trip and was most recently working on the plane. What will fix the problem?

☒ Turning off airplane mode ✔ Correct

☐ Replacing the battery

☐ Hardwiring the computer

☐ Reconnecting the antenna connector

Explanation

This scenario points to airplane mode still being enabled. Laptops can have function keys that toggle airplane mode in addition to toggling via the operating system (OS).

Hardwiring the computer to the network will connect the user to the internet but will not solve the problem of Wi-Fi not working.

Wi-Fi not working is not a sign of a bad battery. A bad battery will be unable to hold a charge for a reasonable amount of time.

If the device had recently been serviced and Wi-Fi stopped working after that service, the antenna connector may have been dislodged or wrongly connected.

Related Content

resources\questions\pt_core1_220_1201_5_4_005.question.xml

Question 12

✕ Incorrect

What format uses a system of mirrors that reflects the illuminated image of the object onto a lens that uses a prism to split the image into its component RGB colors?

☐ Flatbed scanner ✓ Correct

☒ OCR ✕ Incorrect

☐ ADF

☐ SMB

Explanation

A flatbed scanner works by shining a bright light at the object placed on a protective glass surface. This information creates a bitmap file of the object.

The Server Message Block (SMB) protocol means the scan generates as a file on a shared network folder. Each user must have permission to write to the share.

Optical character recognition (OCR) is software that can convert scanned text into digital documents ready for editing.

An automatic document feeder (ADF) passes paper over a fixed scan head. This is a more efficient way of scanning multi-page documents.

Related Content

10.1.13 Scanner Configuration

resources\questions\pt_core1_220_1201_3_7_009.question.xml

Question 13

✓ Correct

What type of printer uses a heating element to create the image on the paper?

- ☐ Laser
- ☐ Inkjet
- ☐ Impact
- ☒ Thermal ✓ Correct

Explanation

A thermal printer is a general term for any device that uses a heating element to create paper images. Thermal printers have a heat-sensitive print that the paper is friction-fed through the print mechanism when in the feed assembly.

Laser printers are the most popular printer technologies for office applications because they are inexpensive, quiet, and fast.

Inkjet printers typically create good-quality color output, such as photo printing. Inkjet is cheap to buy but expensive to run.

The Impact printer strikes an inked ribbon against the paper to leave marks. The most common type is the dot matrix printer.

Related Content

resources\questions\pt_core1_220_1201_3_8_002.question.xml

Question 14

✔ Correct

A company building is under construction and a technician is getting ready to install communication wires. What can a technician use to make the installation simpler?

☒ Plenum ✓ Correct

☐ Fiber

☐ Straight tip

☐ Lucent connector

Explanation

Plenum is a space that is a void in a building designed to carry heating, ventilation, and air conditioning (HVAC) systems. Plenum space is usually a false ceiling or raised floor.

Straight tip (ST) is a type of core fiber optic connector that is a bayonet-style connector and uses a push-and-twist locking mechanism for mostly multi-mode networks.

A lucent connector (LC) is a type of core fiber optic connector that is a small form factor connector with a tabbed push/pull design. LC is similar to a subscriber connector (SC), but the smaller size allows for higher port density.

Fiber cable contains a protective jacket and terminates by a connector. Fiber optic cables fall into two broad categories: single-mode and multi-mode.

Related Content

5.3.4 Cat Standards



5.3.10 Copper Cabling Installation Considerations

resources\questions\pt_core1_220_1201_3_2_011.question.xml

Question 15

✔ Correct

What tool can help diagnose extended read/write times?

☒ SMART ✔ Correct

☐ Pinwheel of death

☐ POST beeps

☐ RAID controller

Explanation

The fixed disk self-diagnostic program called Self-Monitoring, Analysis, and Reporting Technology (SMART) can detect if a drive is close to failure.

A redundant array of individual disks (RAID) controller is what controls the reading and writing from a RAID array.

Power-on self-test (POST) beeps can help diagnose initial problems with hardware, but it does not give any specific information. For example, POST beeps will only indicate, "Something is wrong with xxx.", without specifics.

The pinwheel of death is the Mac OS proprietary crash screen. This crash screen indicates a problem with the operating system.

Related Content

4.2.11 Troubleshoot Drive Reliability and Performance

resources\questions\pt_core1_220_1201_5_2_007.question.xml

Question 16

✔ Correct

What should a software development company utilize to test software applications that must operate under several different OS and resource constraints?

☒ Cross-platform virtualization ✓ Correct

☐ Test development

☐ Resource requirements

☐ Sandbox

Explanation

Cross-platform utilization tests software applications under different OSs and resource constraints and is important to ensure the application is compatible across several different environments.

A software development company should use a sandbox to analyze viruses, worms, and Trojans in a contained environment so that the developers' systems do not become infected.

Test development often uses virtual desktop platforms based on some sort of guest OS hypervisor.

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Related Content

8.1.2 Uses for Virtualization

resources\questions\pt_core1_220_1201_4_1_012.question.xml

Question 17

✔ Correct

What characteristics of Wi-Fi 6 (802.11ax) allow for increased bandwidth and support for modern networks? (Select three.)

- ☒ Orthogonal frequency division multiple access ✔ Correct
- ☒ Uplink and downlink MU-MIMO ✔ Correct
- ☐ Typically supports up to four simultaneous clients
- ☒ Designed to work in both the 2.4 GHz and 5 GHz bands ✔ Correct

Explanation

Wi-Fi 6 works in both the 2.4 GHz and 5 GHz bands. There is emerging development for operation over the 6 GHz band as well (Wi-Fi 6e).

Wi-Fi 6 also adds support for uplink multi-user, multiple input, multiple output (MU-MIMO), along with downlink MU-MIMO. This allows MU-MIMO-capable clients to send data to the access point simultaneously.

Orthogonal frequency division multiple access (OFDMA) can work alongside MU-MIMO to improve client density-sustaining high data rates when more stations connect to the same access point.

Wi-Fi 5 supports up to four simultaneous clients over 5 GHz only, and Wi-Fi 6 can support up to eight clients, giving it better performance in congested areas.

Related Content

-  5.4.3 IEEE 802.11a
-  5.4.4 IEEE 802.11b/g
-  5.4.5 802.11n
-  5.4.7 Wi-Fi 5 and Wi-Fi 6
-  5.4.8 Wi-Fi 7 (802.11be)

resources\questions\pt_core1_220_1201_2_2_007.question.xml

Question 18

— Partial

Which Institute of Electrical and Electronics Engineers (IEEE) wireless standards operate in the 2.4 GHz wireless frequency band? (Select three.)

☒ 802.11g ✓ Correct

☐ 802.11n ✓ Correct

☒ 802.11b ✓ Correct

☒ 802.11a ✗ Incorrect

Explanation

The IEEE 802.11b standard uses the 2.4 GHz frequency band, uses encoding methods inferior to 802.11a, and supports a nominal data rate of just 11 Mbps.

The IEEE 802.11g standard uses the 2.4 GHz frequency band but uses the same encoding mechanism as 802.11a and has an improved 54 Mbps rate.

The IEEE 802.11n standard introduced several improvements to increase bandwidth. It can work over both 2.4 GHz and 5 GHz.

The IEEE 802.11a standard uses the 5 GHz frequency band only. The data encoding method allows a maximum data rate of 54 Mbps.

Related Content

 5.4.2 Frequency Bands

 5.4.4 IEEE 802.11b/g

resources\questions\pt_core1_220_1201_2_2_001.question.xml

Question 19

✔ Correct

An artist looks to achieve greater precision when creating digital drawings. Which input device will help achieve this goal?

☒ Touch pen ✔ Correct

☐ Touch screen

☐ Trackpad

☐ Touchpad

Explanation

Most drawing pads and some touch screens can be used with a touch pen or stylus rather than fingers. A stylus allows for more precise control and can be used for handwriting and drawing.

A trackpad acts as an input device and is a larger-format device attached as a peripheral rather than built into a laptop.

A touch screen is also referred to as a digitizer because of the way it converts analog touch input to digital software instructions. A digitizer sits above a display panel.

A touchpad is a navigation device that is built into laptops and acts as a mouse.

Related Content

9.1.1 Mobile Device Accessories

resources\questions\pt_core1_220_1201_1_2_003.question.xml

Question 20

✔ Correct

A technician troubleshoots a laptop's direct connectivity to the organization's access points connected to an internal network. Troubleshooting starts by removing a screw-down panel to access and examine internal components. Which components does the technician test?

- ☐ Modem
- ☐ Mini PCIe
- ☐ mSATA

☒ Wireless card ✔ Correct

Explanation

A wireless card is an internal component of a laptop that provides Wi-Fi connection capabilities. A wireless card would be seated within the laptop and have an antenna connection.

A modem is a communication device; however, it is not the correct component in this case. The modem would not connect to an access point, as detailed in the scenario.

A mini peripheral component interconnect express (PCIe) card is a form factor that many adapter cards use. These form factors include wireless cards, Bluetooth cards, and more.

A mini serial advanced technology attachment (mSATA) card is a storage device, like a solid-state drive (SSD). An mSATA uses far less power and has a smaller footprint than a traditional SSD drive.

Related Content

9.3.3 RAM and Adapter Replacement



9.3.5 Keyboard and Security Component Replacement

resources\questions\pt_core1_220_1201_1_1_004.question.xml

Question 21

✕ Incorrect

A network engineer must set up an office's new wireless access points (APs). After installing the access points, the engineer must ensure they set up the radio signals correctly and that all office areas receive the most efficient signal possible. What tool will the engineer utilize?

- ☐ Cable tester
- ☐ Loopback plug
- ☐ Wi-Fi analyzer ✓ Correct
- ☒ Toner probe ✕ Incorrect

Explanation

A network engineer can install Wi-Fi analyzer software to record the AP's associated statistics and detect other access points in the vicinity. This can configure the proper signal strength and channels.

A cable tester is a pair of devices attached to each end of a cable. It tests a patch cord or is connected via patch cords to a wall port and patch panel port to test the permanent link.

A toner probe detects the signal and follows the cable over ceilings and through ducts or identifies it from the rest of the bundle. This is strictly for physical cabling.

A loopback plug tests a network interface card (NIC) or switch port. This does not test any wireless metric.

Related Content

5.4.10 Wi-Fi Analyzers

resources\questions\pt_core1_220_1201_2_8_004.question.xml

Question 22

✔ Correct

Which of the following causes the ink in an inkjet printer to break up into small pumps, which give an electrical charge when they leave the print head and can either stick to the paper or fall off?

- ☐ Thermal method
- ☒ Epson design ✓ Correct
- ☐ Transfer roller
- ☐ Imaging drum

Explanation

Epson design, the nozzle contains a piezoelectric element, which changes shape when a voltage is applied. The steady stream is the ink sent through the print head, and the ink that falls off is recycled.

The thermal method is when the ink at each nozzle in the print head is heated, creating a bubble. When the bubble bursts, it sprays ink through the nozzle and draws more ink from the reservoir.

The imaging drum is a round rotating cylinder conditioned by the primary charge roller (PCR) in a laser printer.

The transfer roller applies a positive charge to the underside of the paper in a laser printer.

Related Content

 10.1.11 Printer Security

 10.2.3 Inkjet Printer Imaging Process

resources\questions\pt_core1_220_1201_3_8_007.question.xml

Question 23

✔ Correct

A video gamer is using a product designed to record footage from computer games. What is this called?

- ☐ NIC
- ☐ Video card
- ☒ Capture card ✔ Correct
- ☐ Sound card

Explanation

Capture cards record video input and save it as a movie or streaming media file. Some work with PC games, while others record from a game console or live camera high-definition multimedia interface (HDMI) sources, such as a camcorder or security camera.

Video cards generate the signal to drive a monitor or projector. Low-end graphics adapters are likely to be included with the motherboard chipset or as part of the CPU itself.

A sound card records input from a microphone. Audio playback occurs via speakers or headphones connected to sound cards.

The network interface card (NIC) hardware component on a computer provides a network connection.

Related Content[2.2.14 Capture Cards](#)

resources\questions\pt_core1_220_1201_3_5_003.question.xml

Question 24

✕ Incorrect

A technician uses a U.2 interface on the motherboard that enables storage access and transport to deliver the highest throughput and fastest response time. What component is being used?

☒ SSD ✕ Incorrect

☐ HDD

☐ mSATA

☐ NVMe ✓ Correct

Explanation

NVMe (NVM Express) can be packaged for installation to a PCIe slot as an expansion card or a U.2 or M.2 slot.

A solid-state drive (SSD) uses flash memory technology to implement persistent mass storage. Flash memory performs much better than the mechanical components used in hard drives.

A hard disk drive (HDD) stores data on metal or glass platters coated with a magnetic substance. Desktops use a width of 3.5 inches, and a laptop uses 2.5 inches.

A mini-serial ATA (mSATA) allows a technician to plug an SSD packaged as an adapter card into the motherboard's combined data and power port.

Related Content

3.2.3 Solid-State Drives

resources\questions\pt_core1_220_1201_3_4_003.question.xml

Question 25

✔ Correct

Which of the following allows printers to connect using a Plug and Play that installs the driver automatically?

- ☐ Wireless
- ☒ USB ✔ Correct
- ☐ Ethernet
- ☐ RJ45 port

Explanation

USB connects the device plug to the printer's USB port and the host plug to a free port on the computer.

Wireless has two interfaces: Bluetooth and Wi-Fi. Bluetooth allows access by using the print device control panel to make it discoverable. Then, the Bluetooth page adds the device. Wi-Fi uses two way infrastructure mode and Wi-Fi Direct.

An engineer can configure Ethernet to obtain an Internet Protocol (IP) configuration from a Dynamic Host Configuration Protocol (DHCP) server or they can manually configure this.

RJ45 port is used with an Ethernet network adapter to connect a printer to a computer.

Related Content

10.1.3 Print Device Connectivity

resources\questions\pt_core1_220_1201_3_7_003.question.xml

Question 26

✔ Correct

What is a possible source of data corruption?

- ☐ Loss of internet access
- ☐ Failing RTC battery
- ☒ Sectors or blocks fail ✓ Correct
- ☐ SMART

Explanation

Data corruption can occur when particular sectors or blocks fail, meaning that the system cannot open the files stored in these locations, or they simply disappear.

Although frustrating, loss of internet access will not cause data to become corrupted.

A failing real-time clock (RTC) battery will not corrupt the data. Instead, the main symptom will be incorrect data and time. The RTC battery is known as the complementary metal-oxide-semiconductor (CMOS) battery in older systems.

Self-Monitoring, Analysis, and Reporting Technology (SMART) is a system utility that can help diagnose drive problems, not cause data corruption.

Related Content

4.2.12 Troubleshoot RAID Failure

resources\questions\pt_core1_220_1201_5_2_004.question.xml

Question 27

✔ Correct

This protocol transfers email messages across a network and uses TCP port 25.

☒ SMTP ✔ Correct

☐ Telnet

☐ FTP

☐ SSH

Explanation

Simple Mail Transfer Protocol (SMTP) transfers email messages across a network and uses TCP port 25.

File Transfer Protocol (FTP) makes files available for download across a network. It utilizes two ports. TCP Port 21 establishes the connection between two computers (control connection), and TCP Port 20 is the port the file(s) are transferred across (data connection).

Telnet makes an unsecured connection to the command-line interface of a remote server and uses TCP Port 23.

Secure Shell (SSH) makes a secure connection to the command-line interface of a remote server and uses TCP Port 22.

Related Content

6.3.6 Well-Known Ports

resources\questions\pt_core1_220_1201_2_1_004.question.xml

Question 28

✕ Incorrect

A user finished setting up their new laptop on the latest Windows operating system and is currently setting up print capabilities. They go to pick up the paper from the printer, but everything comes out garbled. What should they troubleshoot first?

- ☐ Clean the rollers
- ☐ Drivers ✓ Correct
- ☒ Creased paper ✕ Incorrect
- ☐ Toner fuser

Explanation

Garbled print typically occurs because of a fault in rendering the print job somewhere in the path between the application, printer driver, page description language, and print device.

Vertical or horizontal line marks that appear in the same place are often due to dirty feed rollers or a damaged or dirty photosensitive drum.

Toner not fused to paper has symptoms of output that smudges easily. This indicates that the fuser needs replacing.

Most paper jams arise because the media (paper or labels) are not suitable for the printer or because a sheet is creased, folded, or not loaded properly in the tray.

Related Content

 10.3.3 Print Quality Issues

 10.3.5 Print Job Issues

resources\questions\pt_core1_220_1201_5_6_001.question.xml

Question 29

— Partial

The 802.11n standard was introduced as an improvement over 802.11a/b/g standards. What characteristics of the 802.11n standard show improved connectivity and performance? (Select three.)

- ☒ It can work only over the 5 GHz band. ✕ Incorrect
- ☒ It features the introduction of MIMO. ✓ Correct
- ☒ The nominal data rate is 64 Mbps per stream or 150 Mbps per stream for a 40 MHz bonded channel. ✓ Correct
- ☐ It can work over both 2.4 GHz and 5 GHz bands. ✓ Correct

Explanation

802.11n can work over both 2.4 GHz and 5 GHz. A technician can implement each band with a separate radio.

Introduced with 802.11n is increased reliability and bandwidth by multiplexing signal streams from 2-3 separate antennas. This technology is known as multiple input multiple outputs (MIMO).

The nominal data rate for 802.11n is 72 Mbps per stream, not 64 Mbps per stream, or 150 Mbps per stream for a 40 MHz bonded channel, and 802.11n access points market using Nxxx designations, where xxx is the nominal bandwidth.

802.11n works over both the 2.4 GHz and 5 GHz bands. 802.11n is also known as Wi-Fi 4.

Related Content

-  5.4.3 IEEE 802.11a
-  5.4.4 IEEE 802.11b/g
-  5.4.5 802.11n
-  5.4.7 Wi-Fi 5 and Wi-Fi 6
-  5.4.8 Wi-Fi 7 (802.11be)

resources\questions\pt_core1_220_1201_2_2_008.question.xml

Question 30

✔ Correct

A company offers a bring your own device (BYOD) program at work. Several users approach the company in an effort to receive reimbursement for changing personal phones to the company's carrier. This will be done by swapping out a subscriber identity module (SIM) card in devices with which network type?

☒ LTE ✔ Correct

☐ HSPA

☐ CDMA

☐ NFC

Explanation

Long-term evolution (LTE) provides an upgrade from CDMA. All 4G and 5G cellular data connections require a SIM card.

Code division multiple access (CDMA) means that the handset is directly managed by the provider, and there is no removable SIM card.

High Speed Packet Access (HSPA) provides improved data rates on GSM networks.

Near-field communications (NFC) allow a mobile device to use a close-range service such as contactless payments.

Related Content

 9.2.6 Enterprise Mobility Management

 20.1.3 Enterprise Mobility Management

resources\questions\pt_core1_220_1201_1_3_011.question.xml

Question 31

✔ Correct

A small shop uses a new point of sale system that allows for touchless payments. The business owner, who is not so tech-savvy, is amazed at the process. As the owner sets up the system, what technology does the system installer mention the payments use?

- ☐ Cellular
- ☐ Bluetooth
- ☐ Wi-Fi
- ☒ NFC ✓ Correct

Explanation

Many mobile devices have near-field communication (NFC) chips built in. NFC allows for very short-range data transmission. Credit card payments can use this technology.

A Bluetooth wireless radio creates a short-range personal area network (PAN) to share data with a PC, connect to a printer, and even connect to automobiles. The connection occurs through what is known as pairing.

Every laptop, smartphone, and tablet supports a Wi-Fi radio. While some automobiles also offer built-in Wi-Fi, this technology is not used to pair two devices.

Cellular data networking means connecting to the internet or services via a device's cellular radio and the phone/system network provider.

Related Content

9.1.10 Near-Field Communication Wireless Connections

resources\questions\pt_core1_220_1201_1_2_008.question.xml

Question 32

✔ Correct

What issue does constant indicator LED activity imply?

- ☐ Connector fault
- ☐ Faulty boot sector
- ☒ Not enough system RAM ✓ Correct
- ☐ Mechanical problem

Explanation

Constantly on light-emitting diode (LED) indicator lights imply disk thrashing. Disk thrashing occurs when there is not enough system random access memory (RAM), so the system continually uses the disk for paging (virtual memory). It could also signify a faulty software process or that malware has infected the system.

A faulty boot sector would display the error message, "Bootable device not found" instead of constantly on LED indicator lights.

If the device does not boot at all, a physical connection in the computer could be faulty.

Loud clicking or grinding noises as the drive is operating indicate mechanical problems with the drive. Therefore, the technician should replace the hard disk drive (HDD) as soon as possible.

Related Content

 4.2.9 Troubleshoot Drive Availability

 4.2.10 Lab: Troubleshoot Drive Availability

resources\questions\pt_core1_220_1201_5_2_001.question.xml

Question 33

✔ Correct

Cloud service providers (CSPs) help organizations save time and money by hosting virtual resources on their hardware. What do CSPs use to ensure efficiency while providing services to customers?

- ☐ Community cloud
- ☐ Public cloud
- ☒ Shared resources ✔ Correct
- ☐ Hybrid cloud

Explanation

Shared resources are correct. To meet availability, scalability, and elasticity requirements, cloud providers must be able to provision and de-provision resources automatically through the pooling of shared resources and virtualization.

Community cloud is incorrect. CSPs provide community clouds to customers using shared resources. A community cloud is a way for organizations to share a private cloud and save money.

The public cloud is incorrect. CSPs provide public clouds to customers using shared resources. CSP offers public cloud on a subscription or pay-as-you-go basis but comes with security and performance risks.

Hybrid cloud is incorrect. CSPs provide hybrid clouds to customers using shared resources. Customers using hybrid clouds can use public cloud resources when their private infrastructure is insufficient.

Related Content

8.2.3 Common Cloud Deployment Models

resources\questions\pt_core1_220_1201_4_2_009.question.xml

Question 34

✔ Correct

A remote user contacts the IT desk to report that they accidentally dropped a mobile device. It is no longer responsive when trying to tap on icons. Which component is likely damaged and is causing the problem?

☒ Digitizer ✓ Correct

☐ Tempered glass

☐ Display panel

☐ Protective glass

Explanation

A touch screen can also be referred to as a digitizer due to the way it converts analog touch input to digital software instructions. The digitizer is sandwiched between a layer of protective glass and the display panel.

A display panel is only a single layer within the screen assembly for a mobile device. A display sits below the digitizer.

Protective glass covers the digitizer on a mobile device. The protective glass and digitizer usually adhere together.

The touchscreen itself is covered by a thin layer of scratch-resistant, shock-resistant tempered glass, such as Corning's Gorilla Glass.

Related Content

2.1.9 Display Components

resources\questions\pt_core1_220_1201_3_1_004.question.xml

Question 35

✔ Correct

Which universal serial bus (USB) 2 connector is square with a beveled top that a technician can use for printers?

- ☐ Type B Micro
- ☐ Type A
- ☒ Type B ✓ Correct
- ☐ Type B Mini

Explanation

Type B connects to large devices. The connector and port are square, with a beveled top.

Type B Mini is a smaller peripheral device connector. This connector was seen on early digital cameras but is no longer widely used.

Type B Micro is an updated connector for smaller devices such as smartphones and tablets. The micro connector is distinctly flatter than the older mini type of connector.

Type A connects to the host and some types of peripheral devices. The connector and port are flat rectangle-shaped. A technician should insert the connector with the USB symbol facing up.

Related Content

 2.1.4 Universal Serial Bus Cables

 2.1.5 USB Standards

resources\questions\pt_core1_220_1201_3_2_001.question.xml

Question 36

✔ Correct

A large real estate agency has acquired a new service from their cloud provider called Desktop as a Service (DaaS). The service will enable the real estate agents to use virtual machines to load their desktops. Which option best describes the service the real estate agency has begun using?

- ☐ Public cloud
- ☐ Private cloud
- ☐ Virtual desktop infrastructure (VDI) on-premises
- ☒ Virtual desktop infrastructure (VDI) in the cloud ✔ Correct

Explanation

Virtual desktop infrastructure (VDI) in the cloud is often referred to as Desktop as a Service (DaaS) and enables users to load their desktops from a virtual machine hosted in the cloud.

Virtual desktop infrastructure (VDI) on-premises means that the virtualization server hosting the virtual desktops has been provisioned on the same local network as the clients.

A public cloud is a cloud service offered over the internet by cloud providers to customers who share resources with other customers and pay on a subscription or pay-as-you-go basis.

Private cloud is completely private to and owned by the organization that pays for the service. The private cloud does not provide VDI services alone.

Related Content

resources\questions\pt_core1_220_1201_4_2_013.question.xml

Question 37

✔ Correct

A manager needs to electronically sign many documents while using a laptop. Precision is not required, and there is very little money in the budget to spend. What does the manager choose to use as a solution?

☐ Drawing pad☐ Touch pen☐ Trackpad☒ Touchpad ✔ Correct**Explanation**

A touchpad is a navigation device that is built into laptops and acts as a mouse. Using a finger with the touchpad can cost-efficiently accomplish the manager's needs.

Most drawing pads and some touch screens can be used with a touch pen or stylus rather than fingers. A stylus allows for more precise control and can be used for handwriting and drawing.

A trackpad acts as a mouse and is a larger-format device attached as a peripheral rather than built into a laptop. It may also be used on desktops.

A drawing pad refers to a large-format touch device attached as a peripheral. These are also called graphics tablets, as they are used in digital art applications.

Related Content

9.1.1 Mobile Device Accessories

resources\questions\pt_core1_220_1201_1_2_007.question.xml

Question 38

✕ Incorrect

A growing company is looking to move away from an outside vendor for payment processing and would like to do it in-house. As a part of securing the data on these computers, what foundational infrastructure solution should the company implement to keep this data separate from the rest of the company's internal network?

- ☐ DNS (A) record
- ☒ DHCP scope ✕ Incorrect
- ☐ Virtual LAN ✓ Correct
- ☐ Virtual private network

Explanation

A virtual LAN (VLAN) is a logical network segment using a feature of managed switches to segregate network traffic. This would be the first step to separating network traffic for the given scenario.

A virtual private network (VPN) enables mostly remote hosts to connect to the LAN without being physically installed at the site.

In the Domain Name System (DNS), an A record resolves a hostname to an IPv4 address. This would not provide any added security.

A dynamic host configuration protocol (DHCP) scope is a range of consecutive IP addresses in the same subnet that a DHCP server can lease to clients. The company may not use this if all the hosts have static addresses.

Related Content[6.4.6 Virtual LANs](#)[resources\questions\pt_core1_220_1201_2_4_004.question.xml](#)

Question 39

✔ Correct

IT systems administrators for a company have been busy configuring a fleet of new smartphones for users. Instructions that the administrators follow to configure the devices include using a host address and entering a domain as an optional parameter. Which corporate email service does the company use?

- ☐ Google Workspace
- ☒ Microsoft Exchange ✓ Correct
- ☐ Gmail
- ☐ Microsoft 365

Explanation

To manually configure a Microsoft Exchange account, parameters such as a host address as well as the choice of whether to use transport layer security (TLS) are required. There is often also an optional field for a domain, but this is usually left blank.

A Microsoft digital identity is used to access cloud subscriptions for the Office productivity software suite and the OneDrive cloud storage service, which are known as Microsoft 365. The configuration only uses a username and password.

With Google Workspace, a Google account grants access to Google's Workspace productivity software and other Google services. This configuration does not use a domain.

A Gmail account is a free account used with the Google Workspace landscape. Configuration does not use a domain.

Related Content

 9.2.6 Enterprise Mobility Management

 20.1.3 Enterprise Mobility Management

resources\questions\pt_core1_220_1201_1_3_003.question.xml

Question 40

✔ Correct

A printer allows connection to a network through the infrastructure mode. Which of the following connections is the printer most likely using when referring to an infrastructure mode connection?

- ☐ Duplex Unit
- ☐ Ethernet
- ☒ Wireless ✓ Correct
- ☐ USB

Explanation

Wireless has two interfaces: Bluetooth and Wi-Fi. Bluetooth allows access by using the print device control panel to make it discoverable. Then, the Bluetooth page adds the device.

USB connects the device plug to the printer's USB port and the host plug to a free port on the computer.

An engineer can configure Ethernet to obtain an Internet Protocol (IP) configuration from a Dynamic Host Configuration Protocol (DHCP) server or they can manually configure this.

Engineers use duplex units for double-sided printing and a finisher unit for folding, stapling, and hole punching.

Related Content

10.1.3 Print Device Connectivity

resources\questions\pt_core1_220_1201_3_7_002.question.xml

Question 41

✕ Incorrect

A holiday decor store typically runs its cloud infrastructure privately, but its servers get bogged down by transactions during the peak holiday season. To mitigate any revenue losses, the store must outsource processing power to an outside cloud service provider to help support the increased sales volume. What cloud deployment model does the holiday decor store utilize?

☒ Public cloud ✕ Incorrect

☐ Private cloud

☐ Community cloud

☐ Hybrid cloud ✓ Correct

Explanation

Hybrid cloud deployment is what the holiday decor store utilizes because it allows the store to utilize its private cloud infrastructure during non-peak sales times and then incorporate a public cloud for support during peak sales times.

Private cloud is incorrect because it does not allow the decor store to outsource resources during peak sales.

The public cloud is incorrect because it does not allow the decor store the opportunity to use its internal cloud infrastructure.

Community is not the best answer because the decor store does not want to share resources with other organizations year-round.

Related Content

8.2.3 Common Cloud Deployment Models

resources\questions\pt_core1_220_1201_4_2_003.question.xml

Question 42

✔ Correct

A systems engineer is doing an internal audit of their virtual environment before an annual inspection and is making sure that all the virtual machines are up to date on patches. What could the systems engineer consult to verify which patches need to be on each machine?

- ☐ Legacy software/OS
- ☐ Test development
- ☐ Resource requirements

☒ Security requirements ✔ Correct

Explanation

Security requirements are correct because patching is a part of ensuring a system is secure against the latest vulnerabilities.

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Test development often uses virtual desktop platforms that are based on some sort of guest OS hypervisor.

Legacy software applications and operating systems may not perform well if a user's system has been upgraded.

Related Content

8.1.5 Virtualization Security Requirements

resources\questions\pt_core1_220_1201_4_1_007.question.xml

Question 43

✕ Incorrect

What might be diagnosed by no LED status indicators on a drive?

- ☐ Keyboard issue
- ☐ Bad sectors
- ☒ Corrupted boot sector ✕ Incorrect
- ☐ Faulty disk ✓ Correct

Explanation

No LED status indicator lights could indicate that the whole system might not be receiving power or that the individual disk could be faulty.

A read/write error can indicate faulty sectors in a hard disk drive (HDD). Likewise, a power failure or a mechanical fault can damage a sector. If the technician runs a test utility, such as the `chkdsk` command, and `chkdsk` locates more bad sectors each time the technician runs the test, it is a sign that the disk is about to fail.

During the power-on self-test (POST), three long beeps will indicate a keyboard issue.

The error message, 'Boot device not found,' would indicate a corrupted boot sector.

Related Content

4.2.9 Troubleshoot Drive Availability



4.2.10 Lab: Troubleshoot Drive Availability

resources\questions\pt_core1_220_1201_5_2_010.question.xml

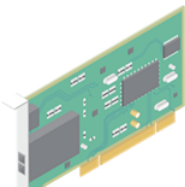
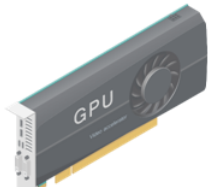
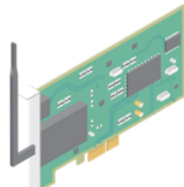
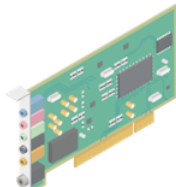
Question 44

— Partial

Instructions

Based on the scenario, use the **dropdown selectors** to identify the correct expansion card slot for the card listed, then use the **checkboxes** to select the other component(s) that are NOT integrated into the motherboard.

You are presented with 4 expansion cards as follows: a RJ45 NIC Card, a Video Card, a Wireless NIC Card, and a Sound Card. You must use the single dropdown selector associated with each card image to select the correct motherboard slot to place the card in. Following the cards, you are asked to select the set of components that are NOT integrated into the motherboard. There are 6 associated checkboxes for selecting the components.

RJ45 NIC Card	Video Card	Wireless NIC Card	Sound Card
PCIe x8 (v1.x) ☒	PCIe x16 (v1.x) ☒	PCIe x1 (v1.x) ☒	PCIe x4 (v1.x) ☒
			

☒ You got **6 out of 10** correct. Slot for RJ45 NIC Card answered PCIe x8 (v1.x), incorrect Less

You can determine which slot is appropriate for each card based on the connector requirements per slot.

The RJ45 card in this scenario is connected to the traditional PCI slot. This can be identified by segments in the connector pins on the bottom side of the card. A PCI slot has a small section, a break, and then a large section on its connector pins. The PCI has a slightly larger initial segment before the long portion on the connector pin compared to a PCIe x16.

The video card (or graphics adapter), which uses a PCIe x16 (v1.x) expansion card, generates a signal to drive a monitor or projector. The PCIe x16 is differentiated by the smallest initial segment of connector pins, followed by a break, then the long portion of the 16-lane configuration. PCIe x16 can be identified by the slightly smaller initial section compared to the traditional PCI card.

A Wireless NIC card, which uses a PCIe x1 (v1.x) expansion card, has a Wi-Fi adapter added instead of RJ45 ports and connects to a wireless

Related Content

pbq2-core1.question.fex

Question 45

✔ Correct

An automobile allows for connecting a smartphone or tablet for entertainment and audio purposes. Pairing the two occurs with which technology?

☒ Bluetooth ✔ Correct

☐ Hotspot

☐ NFC

☐ Wi-Fi

Explanation

A Bluetooth wireless radio creates a short-range personal area network (PAN) to share data with a PC, connect to a printer, and even connect to automobiles. The connection occurs through what is known as pairing.

Many mobile devices have near-field communication (NFC) chips built in. NFC allows for very short-range data transmission (up to about 20 cm/8 in.) to activate a receiver chip in a contactless reader.

A smartphone or tablet can be configured as a personal hotspot to share its cellular data connection with other computer devices.

Every laptop, smartphone, and tablet supports a Wi-Fi radio. While some automobiles also offer built-in Wi-Fi, this technology is not used to pair two devices.

Related Content

 9.1.2 Mobile Device Wired Connection Methods

 9.1.9 Bluetooth Wireless Connections

resources\questions\pt_core1_220_1201_1_2_002.question.xml

Question 46

✔ Correct

A user has accidentally put too much paper in the feed. What possible issues might this cause? (Select two.)

- ☐ Misaligned binding
- ☐ Backed-up print queue
- ☒ Staple jams ✔ Correct
- ☒ Hole punching issues ✔ Correct

Explanation

An excessive number of sheets is also the primary cause of staple jams. Remove the staple cartridge, and release the catch at the end to allow the removal of stuck staples.

The main issue with hole punching is exceeding the maximum number of sheets. Make sure print jobs are sent in batches of less than the maximum permissible sheet count.

The paper size and orientation must be set correctly for the print job or the finishing/binding will be aligned to the wrong edge.

A backed-up print queue means that there are multiple prints pending but not printing. This might occur because the print device is offline or out of paper or ink/toner.

Related Content

10.3.4 Finishing Issues

resources\questions\pt_core1_220_1201_5_6_004.question.xml

Question 47

✕ Incorrect

A vast majority of rural residents in the United States still use this internet connection type that utilizes the public switched telephone network (PSTN).

- ☐ Satellite
- ☐ DSL ✓ Correct
- ☐ WISP
- ☒ Wi-Fi ✕ Incorrect

Explanation

A digital subscriber line (DSL) uses the higher frequencies available in copper telephone lines as a communications channel. The two major "flavors" of this type of connection are asymmetrical and symmetrical.

Various satellite internet solutions are available to rural residents, but they do not utilize the PSTN infrastructure.

A wireless internet service provider (WISP) uses ground-based long-range fixed access wireless technology. The WISP installs and maintains a directional antenna to bridge the customer's network and the service provider.

Wi-Fi is the brand name for the IEEE 802.11 standards used to implement wireless local area networks (WLANs).

Related Content

 6.1.1 Internet Connection Types and Modems

 6.1.2 Digital Subscriber Line Modems

resources\questions\pt_core1_220_1201_2_7_005.question.xml

Question 48

✔ Correct

What would cause a central processing unit (CPU) to make only minor changes to the CPU internals that significantly improve performance and increase speed?

- ☐ Multicore
- ☒ Multithreading ✓ Correct
- ☐ Multisocket
- ☐ Single core

Explanation

Multithreading is also known as HyperThreading by Intel. A thread is a stream of instructions generated by a software application. This provides multiple threads of execution at once, supported by the operating system.

Multicore is a design that allows each core to communicate with the others, allowing the processing tasks to be divided and assigned properly. The multicore acts as a single unit.

Multisocket offers multiple sockets and supports two processors for the CPU.

Single core CPU has a single execution unit and set of registers implemented on a single package.

Related Content

resources\questions\pt_core1_220_1201_3_5_006.question.xml

Question 49

✔ Correct

A system administrator implemented a physical device that expedites cryptographic operations and manages cryptographic keys. What is this device the system administrator implemented?

☐ Boot options☐ TPM☐ Secure boot☒ HSM ✔ Correct**Explanation**

A hardware security module (HSM) is a secure USB key or thumb drive used to store cryptographic material.

A trusted platform module (TPM) is a specification for hardware-based storage of digital certificates, cryptographic keys, and hashed passwords.

Boot options define how the system firmware searches devices for a boot manager. It is one of the most important parameters in a system setup.

Secure boot is a unified extensible firmware interface (UEFI) feature designed to prevent a computer from being hijacked by malware. Under secure boot, the computer firmware configuration has cryptographic keys that can identify trusted code.

Related Content[4.1.8 Trusted Platform Modules](#)

resources\questions\pt_core1_220_1201_3_5_010.question.xml

Question 50

✔ Correct

What features of the cloud allow multiple users to simultaneously access content and work collaboratively while tracking edits by each user?

- ☐ Shared resources
- ☐ High availability
- ☒ File synchronization ✓ Correct
- ☐ Community cloud

Explanation

File synchronization is an advantage of cloud storage that replicates files between data centers and different devices such as PCs and smartphones. Cloud storage also allows multiple users to access content and work collaboratively simultaneously.

To meet availability, scalability, and elasticity requirements, cloud providers must be able to provision and de-provision resources automatically through the pooling of shared resources and virtualization.

A community cloud is a cloud environment deployed for shared use by cooperating tenants. These tenants usually have similar security and operating concerns.

High availability means that the service experiences very little downtime. Downtime can occur because of scheduled maintenance and unexpected outages.

Related Content

-  2.2.4 Motherboard Storage Connectors
-  3.2.3 Solid-State Drives
-  3.2.11 Optical Drives
-  4.2.13 Troubleshooting Problems with Storage Drives & RAID Arrays
-  8.2.7 Cloud File Storage
-  12.4.2 Storage
-  15.2.2 Windows Domains and Active Directory
-  21.1.4 Backup Media Requirements

resources\questions\pt_core1_220_1201_4_2_014.question.xml

Question 51

✓ Correct

A basic firewall configuration has rules, referred to as network access control lists (ACLs). What are the basic properties needed for a firewall ACL to implement the desired security features configured? (Select three.)

☒ Protocol ✓ Correct

☐ MAC address

☒ Block/Allow property ✓ Correct

☒ Source and destination network/device ✓ Correct

Explanation

The ACL requires source and destination network or device for the firewall to know what networks and devices are allowed to communicate with each other.

A protocol needs to be defined in an ACL to narrow the scope of network traffic managed by the ACL.

The network technician can set the ACL to either allow or block specific protocols and traffic between various sources and destination networks and devices.

Firewalls typically manage network traffic at a different layer in the Open Systems Interconnection (OSI) model, and MAC addresses are usually not found in traditional firewall ACLs.

Related Content



6.1.10 Firewalls

resources\questions\pt_core1_220_1201_2_5_008.question.xml

Question 52

✔ Correct

A user complains that their phone is not responding to touch. What is the likely culprit?

- ☐ Damaged port
- ☐ Faulty battery
- ☒ Faulty digitizer ✓ Correct
- ☐ Overheating phone

Explanation

A touch screen can also be referred to as a digitizer due to the way it converts analog touch input to digital software instructions. A faulty digitizer will cause the touch screen to stop working.

When a phone no longer holds a charge for a reasonable amount of time, the user should suspect a faulty battery. The issue can be a damaged battery, or it is simply at the end of its life.

The main symptom of overheating will be that the device is almost too hot to touch. As well, overheating can sometimes cause the phone to turn off.

A damaged port will have intermittent or no connectivity. This issue can occur when charging or using other accessories, such as headphones.

Related Content

9.4.4 Screen and Calibration Issues

resources\questions\pt_core1_220_1201_5_4_007.question.xml

Question 53

✔ Correct

A user has been working with a large transportation company for 30 years and has never seen a double image before. However, when they print the latest report on the new printer, they see a double image on the print. What should they do first?

- ☒ Clean the drum. ✔ Correct
- ☐ Check the toner installation location.
- ☐ Check for mechanical problems.
- ☐ Use a toner vacuum.

Explanation

Double/echo images are a sign that the photosensitive drum has not been cleaned properly. Try printing a series of different images and see if the problem resolves itself. If not, replace the drum/toner cartridge.

For an incorrect chroma display, prints come out in the wrong color. Ensure that the toner cartridges have been installed in the correct location.

A loud or grinding noise, or any sort of clicking sound, is a sign of a mechanical problem.

Speckling on output indicates loose toner may be getting onto the paper. Clean the inside of the printer using an approved toner vacuum.

Related Content

10.3.3 Print Quality Issues

resources\questions\pt_core1_220_1201_5_6_003.question.xml

Question 54

✕ Incorrect

What is the most important output voltage since it is the most heavily used by system components like an HDD or optical drive to drive the motors that spin the disk platters or optical disc?

☐ 12V ✓ Correct☒ 2.2V ✕ Incorrect☐ 5V (standby)☐ 5V**Explanation**

12V offers a maximum load of 33 and a maximum output of 396. The distribution refers to how much power generates over each rail. An HDD or optical drive motor uses this voltage to spin the platters of the disk drive or optical disc.

5V offers a maximum load of 20 and a maximum output of 130. The output for 5V and 3.3V has a combined limit.

2.2V is not an option, but 3.3V offers a maximum load of 20 and a maximum output of 130. The 3.3V has the same output as 5V.

5V (standby) has a maximum load of 2.5 and a maximum output of 12.5.

Related Content 3.1.3 Wattage Rating 3.1.4 Power Supply Connectors 3.1.5 20-pin to 24-pin Motherboard Adapter

resources\questions\pt_core1_220_1201_3_6_005.question.xml

Question 55

✔ Correct

While planning a new virtual environment, a systems engineer noted that one virtual machine (VM) would run a Windows OS and store two snapshots at all times. The VM will need to have at least 80 GB in its hard disk drive for the VM to function properly. What is the systems engineer considering when planning this aspect of the new virtual environment?

☒ Resource requirements ✓ Correct

☐ Security requirements

☐ Legacy software/OS

☐ Test development

Explanation

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Security requirements are incorrect because the amount of central processing units (CPUs) and random access memory (RAM) a system needs will not impact the security of a system.

Legacy software applications and operating systems may not perform well if a user's system has been upgraded.

Test development often uses virtual desktop platforms based on some sort of guest OS hypervisor.

Related Content

resources\questions\pt_core1_220_1201_4_1_006.question.xml

Question 56

✔ Correct

A remote support technician is speaking with a new rural business client who has issues with their internet connection. As the remote technician gathers information for a local technician to visit the site, the user mentions that the local tech referred to their connection as the "local loop" and the "last mile." What device would the remote technician conclude that they use to connect to the internet?

- ☐ Optical network terminal
- ☒ DSL modem ✔ Correct
- ☐ Wi-Fi access points
- ☐ Cable modem

Explanation

A digital subscriber line (DSL) uses higher frequencies in copper telephone lines from the "last mile" or "local loop" of plain old telephone systems (POTS) as a communications channel. A DSL modem interfaces between telephone lines and local networking devices, providing routing and internet connection.

A cable modem is a cable-internet-access digital modem using a coaxial connection to the service provider's fiber-optic core network. This connection is not typical in rural areas.

Wi-Fi access points provide connectivity for devices to the internet; they depend on local routing for internet connection. Modems typically do this routing.

An optical network terminal (ONT) converts between optical and electrical signaling to facilitate full fiber internet connection types. This is not typical in rural communities.

Related Content

 6.1.1 Internet Connection Types and Modems

 6.1.2 Digital Subscriber Line Modems

resources\questions\pt_core1_220_1201_2_5_002.question.xml

Question 57

✔ Correct

An organization rolls out a security update that utilizes either a contactless smart card reader or a fingerprint scanner for login to mobile devices. What type of security implementations is the organization allowing for user logins? (Select two.)

- ☐ Bluetooth
- ☐ Wi-Fi 6
- ☒ Near-field ✔ Correct
- ☒ Biometrics ✔ Correct

Explanation

A biometric sensor allows users to record a template of a feature of their body that is unique to them. On a laptop, this is typically a fingerprint scanner.

A near-field communication (NFC) scanner is an authentication mechanism. It may also pair peripheral devices or establish a connection to a smart device.

Smart card readers do not use Bluetooth and typically connect peripheral devices to PCs and mobiles. Smart card readers also share data between two systems.

Wi-Fi 6 works in both the current 2.4 GHz and 5 GHz bands and adds support for a new 6 GHz frequency band. However, it is not applicable in this scenario.

Related Content

 9.1.10 Near-Field Communication Wireless Connections

 9.3.5 Keyboard and Security Component Replacement

resources\questions\pt_core1_220_1201_1_1_002.question.xml

Question 58

✔ Correct

Each Ethernet network interface card (NIC) port has a unique hardware/physical identifier used for network routing and switching. What is this unique hardware or physical identifier?

- ☐ VLAN ID
- ☐ Internet protocol address
- ☒ Media access control address ✔ Correct
- ☐ Service set identifier

Explanation

Every NIC port has a unique hardware/physical address called the media access control (MAC) address. Each frame of Ethernet data identifies the source MAC address and destination MAC address in a header field.

The core protocol in TCP/IP is the Internet Protocol (IP), which provides network and host addressing and packet forwarding between networks. This is not specific to the hardware.

A service set identifier (SSID) is a character string that identifies a particular wireless LAN (WLAN).

VLAN IDs are on network switch ports for traffic management and network segmentation. These are assigned by software and not native to the hardware.

Related Content[6.1.9 Routers](#)

resources\questions\pt_core1_220_1201_2_5_003.question.xml

Question 59

— Partial

Instructions

Based on the scenario, drag the **troubleshooting task** to the appropriate display issue to help resolve it, then use the **dropdown selectors** to complete the statements or answer the questions.

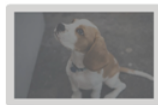


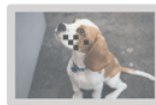
Image appears very dim.

Adjust brightness/contrast



Color glitches. Purple or green horizontal lines.

Verify source in OSD menu



Stuck or dead pixels.

Try to reactivate via touch or software



You will use the first three statements about video to the statement follows:
NK. 2. A flicker takes a "pop" :

To minimize cabling, audio and video can be sent over a single cable using HDMI or:

DisplayPort



A flickering image could also be caused by the display's backlight or what other component starting to fail?

Power supply



You got 6 out of 9 correct.

Less

If an image appears very dim, adjust the display's brightness/contrast settings.

If a display has color glitches like green or purple horizontal lines, check that display cables and ports are fully connected and working properly. If needed, replace them.

If there are stuck or dead pixels, you may be able to reactivate the pixels by gently touching them or using special software, but likely you'll need to replace the display.

If a display has image burn-in, you may need to replace the display. To avoid burn-in, use automatic screensavers or turn off the display. This is mostly only an issue for older CRT or plasma displays.

Related Content

pbq3-core1.question.fex

Question 60

✕ Incorrect

What is the typical use for virtual desktop platforms based on guest OS hypervisor?

- ☐ Cross-platform virtualization
- ☒ Legacy software/OS ✕ Incorrect
- ☐ Testing and development ✓ Correct
- ☐ Resource requirements

Explanation

Desktop virtual platforms, usually based on guest OS, are typically used for testing and development.

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Cross-platform utilization tests software applications under different OSs and resource constraints and is important to ensure the application is compatible across several different environments.

Legacy software applications and operating systems may not perform well if a user's system has been upgraded.

Related Content

8.1.2 Uses for Virtualization

resources\questions\pt_core1_220_1201_4_1_009.question.xml

Question 61

✔ Correct

A consumer new to iOS devices is unsure of the type of cable required to charge their new iPhone 14. The consumer would like an extra cable to keep in a home office. What type of proprietary cable does the consumer look to purchase?

☒ Lightning ✔ Correct

☐ USB

☐ Serial

☐ LCD

Explanation

iPhones, from Generation 5 through 14, and older iPad devices use the proprietary Lightning connector. Some of the latest iPhone and iPad models, such as the iPhone 15 and newer, and iPad Pro, use USB-C.

Modern Android-based smartphones and tablets use the USB-C connector for wired peripherals and charging. The Micro-B USB and Mini-B connector form factors are only found on old devices.

Serial is one of the oldest and simplest computer interfaces. While not many mobile devices have hardware serial ports, the software serial port is often used for programming and connectivity.

Most mobile devices use a flat-panel screen technology based on a type of liquid crystal display (LCD). A liquid crystal display is not a cable type.

Related Content

9.1.2 Mobile Device Wired Connection Methods

resources\questions\pt_core1_220_1201_1_2_006.question.xml

Question 62

✔ Correct

A technician suspects a mobile device to have damage caused by liquid. What should the technician do first?

- ☒ Power off the device. ✓ Correct
- ☐ Replace the battery.
- ☐ Request a warranty replacement.
- ☐ Replace the screen.

Explanation

The technician should power off the device to avoid further damage and remove any excess liquid. Next, the technician must disassemble the device to dry the internal components when exposed to liquid.

The technician will need to replace the battery when a device has liquid damage; however, this is not the first step.

Sometimes liquid damage can result in seeing liquid under the screen. This does not necessarily mean that the screen needs replacing; sometimes, it just needs to dry out.

Depending on the warranty policy, a warranty replacement for water damage to a mobile device may be appropriate. However, the first step will be to power down the device before moving forward.

Related Content

9.4.2 Hardware Failure Issues

resources\questions\pt_core1_220_1201_5_4_006.question.xml

Question 63

✕ Incorrect

A laptop user at a company asks the IT department for computer equipment. In addition to an external monitor, the user would like to take advantage of an expansion card. What does IT purchase for the user?

- ☐ Laptop ports
- ☒ Port replicator ✕ Incorrect
- ☐ Serial interface
- ☐ Docking station ✓ Correct

Explanation

A docking station is a sophisticated port replicator. When docked, a portable computer can function like a desktop machine or use additional features, such as a full-size expansion card.

A port replicator either attaches to a special connector on the back or underside of a laptop or is connected via USB. It provides a full complement of ports for devices such as keyboards, monitors, mice, and network connections.

Serial is one of the oldest and simplest computer interfaces. While not many mobile devices have hardware serial ports, the software serial port is often used.

Laptops ship with standard wired ports for connectivity. The ports are usually arranged on the left and right edges.

Related Content

9.1.3 Port Replicators and Docking Stations

resources\questions\pt_core1_220_1201_1_2_004.question.xml

Question 64

✔ Correct

A user calls to report that they cannot connect to the internet and various other internal resources from their laptop. After a technician asks them what IP address their Windows laptop has, they report it has an address of 169.254.0.115. What problem is the user's laptop having?

- ☐ The laptop has a Class A private address.
- ☐ The laptop has a misconfigured static IP address.
- ☐ The laptop has a misconfigured default gateway address.
- ☒ The laptop cannot get an address from the DHCP server. ✔ Correct

Explanation

Automatic private IP addressing (APIPA) is a mechanism for Windows hosts to obtain an address automatically that cannot contact a DHCP server to revert to using an address from the range 169.254.x.y. This is also called a link-local address.

Using static addressing requires an administrator to visit each computer to enter the configuration information for that host manually. This IP would likely not be statically assigned.

10.0.0.0 to 10.255.255.255 is a Class A private address range.

The default gateway parameter is the IPv4 address of a router. If the laptop cannot contact a DHCP server, it will not get a default gateway at all.

Related Content

 6.2.12 Static Versus Dynamic Host Address Configuration
resources\questions\pt_core1_220_1201_2_6_003.question.xml

Question 65

✔ Correct

A startup company does not have the funds or the time to buy and configure its computer resources, so it pays a service provider to deploy IT resources such as servers, load balancers, and storage components. What service is the startup company paying for?

- ☐ Platform as a service (PaaS)
- ☐ Software as a service (SaaS)
- ☒ Infrastructure as a service (IaaS) ✔ Correct
- ☐ Shared resources

Explanation

Infrastructure as a service (IaaS) is correct. IaaS is a means of provisioning IT hardware and the internet links they require, as needed, from a service provider's data center.

Software as a service (SaaS) is incorrect. SaaS is a means of purchasing software licenses for a given number of users and does not include IT hardware resources.

Platform as a service (PaaS) is incorrect. PaaS is incorrect because the question does not mention that the company needs IT hardware resources and multi-tier web applications and database platforms on top of the provided IT hardware resources.

Shared resources are not a service that can be purchased. Service providers can provide services using shared resources to lower the cost to the customer.

Related Content

8.2.5 Common Cloud Service Models

resources\questions\pt_core1_220_1201_4_2_005.question.xml

Question 66

✔ Correct

A gamer wants to connect a headset to a new gaming laptop. What connection options does the gamer have? (Select three.)

☐ Nib☒ USB ✔ Correct☒ Bluetooth ✔ Correct☒ 3.5 mm jack ✔ Correct**Explanation**

A Bluetooth wireless radio creates a short-range personal area network (PAN) to share data with a PC, connect to a printer, and connect to peripherals such as headsets. The connection occurs through what is known as pairing.

A USB connection is a popular connection for many computer peripherals. This includes mice, keyboards, headsets, gamepads, and more.

A 3.5 mm jack is a standard headphone/speaker jack size found in audio headsets, headphones, microphones, and more.

A nib is not a connector type. Touch pens designed for use with drawing pads have removable and changeable nibs for use as different pen/brush types.

Related Content 9.1.2 Mobile Device Wired Connection Methods 9.1.9 Bluetooth Wireless Connections

resources\questions\pt_core1_220_1201_1_2_009.question.xml

Question 67

✔ Correct

Which of the following would a technician use in small, low-power motherboards?

☒ ITX ✔ Correct

☐ ATX

☐ SAN

☐ BTX

Explanation

Information technology eXtended (ITX) has two designs: the mini-ITX and the mobile-ITX. The mini-ITX is 6.7 inches square with one expansion slot. Technicians can mount most mini-ITX boards in ATX cases.

An advanced technology eXtended (ATX) specifies the standard form factor for most desktop PC motherboards.

A balanced technology eXtended (BTX) improves ATX since it has better airflow.

A storage area network (SAN) is more flexible and reliable than using local disks on each server machine. It provides access to a configurable group of storage devices that application servers can use.

Related Content

2.2.3 Motherboard CPU and System Memory Connectors



2.2.7 Motherboard Form Factors

resources\questions\pt_core1_220_1201_3_5_011.question.xml

Question 68

✔ Correct

How can a technician diagnose capacitor swelling?

- ☒ Physical inspection ✔ Correct
- ☐ Computer that is hot to the touch
- ☐ System diagnostics
- ☐ Slow computer

Explanation

Capacitors are barrel-like components that regulate the flow of electricity to system chips. Swollen, bulging, or emitting any residue indicates damaged or failed capacitors. The only sure way to diagnose this is visual.

A computer that is hot to the touch can indicate overheating, not capacitor swelling.

A sluggish computer can result from many different problems and is not a diagnostic indicator of capacitor swelling.

System diagnostics can help diagnose software and some hardware problems, and it can also point toward capacitor swelling. However, a physical inspection will still be necessary.

Related Content

4.3.1 Troubleshoot Component Issues

resources\questions\pt_core1_220_1201_5_1_010.question.xml

Question 69

✔ Correct

Remote wipe, locator apps, device encryption, remote backup, and antivirus software fall under which category of requirements for virtualized environments?

- ☐ Test development
- ☐ Legacy software/OS
- ☐ Resource requirements
- ☒ Security requirements ✔ Correct

Explanation

Security requirements are correct because all the things listed in the question refer to profile security requirements established for any system.

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Test development often uses virtual desktop platforms that are based on some guest OS hypervisor.

Legacy software applications and operating systems may not perform well if a user's system has been upgraded.

Related Content

8.1.5 Virtualization Security Requirements

resources\questions\pt_core1_220_1201_4_1_010.question.xml

Question 70

✔ Correct

A computer has multiple hard drives installed, but the second hard drive does not show up in Windows Explorer. What is a possible problem?

- ☐ Overheating
- ☐ Corrupted boot sector
- ☐ Bad sectors
- ☒ Cable or connector fault ✔ Correct

Explanation

Suppose the system boots, but a second fixed disk or removable drive does not appear in File Explorer. The technician should first check that the system has initialized and formatted with a partition structure and file system. If the system does not detect the disk by a configuration tool, such as Windows Disk Management, the technician will suspect that there is a hardware or cable/connector fault.

The computer booted up correctly, so a corrupted boot sector is not the problem.

Overheating is generally detected by the computer becoming hot to the touch.

Bad sectors occur when parts of a hard disk drive (HDD) fail, resulting in read/write failures and not the inability to see a hard drive in the OS.

Related Content

-  4.2.5 Troubleshoot Boot Issues
-  4.2.6 Lab: Troubleshoot Boot Issues
-  4.2.9 Troubleshoot Drive Availability
-  4.2.10 Lab: Troubleshoot Drive Availability
-  14.4.8 Troubleshoot Boot Issues

resources\questions\pt_core1_220_1201_5_2_012.question.xml

Question 71

— Partial

Instructions

Based on the scenario, drag the **storage devices** in order from least volatile to most volatile, then use the **radio selectors** to identify if that storage device requires power to store data.

You are presented with a scale of 5 storage devices arranged from least volatile to most volatile. The storage devices are not identified. You must use the 5 associated drop zones to identify each storage device on the scale. Next, you must use the associated set of 2 radio selectors to select if that storage device requires power to store data, or not.

Least Volatile ← → Most Volatile

Storage Device mSATA ✓	Storage Device USB Stick ✗	Storage Device 2.5" HDD ✗	Storage Device CPU Cache ✗	Storage Device RAM ✗
Does device require external power to	Does device require external power to	Does device require external power to	Does device require external power to	Does device require external power to

✗ You got 2 out of 10 correct.

Less

If memory is volatile, you cannot store data without a power supply. The storage devices are stated from least to most volatile in the following order: mSATA form factor SSD, 2.5" form factor HDD, USB Stick (Removable storage), RAM, and finally CPU cache. USB Sticks (Removable storage), 2.5" form factor HDDs, mSATA form factor SSDs are mass storage devices that are slower but do not require power to store data. CPU cache and RAM are fast but require power to store data.

Related Content

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Question 72

✔ Correct

A network technician installs wireless access points (APs) in an office and would like to use the power over Ethernet (PoE) features available. The specific model of APs available requires a draw of 20W with a minimum current of 450 mA. Which Institute of Electrical and Electronics Engineers (IEEE) PoE standard best suits this requirement?

- ☐ 802.11ac
- ☐ 802.3af
- ☒ 802.3at ✓ Correct
- ☐ 802.3bt

Explanation

802.3at (PoE+) allows powered devices to draw up to about 25 W, with a maximum current of 600 mA.

802.3bt (PoE++ or 4PPoE) supplies up to about 51 W (Type 3) or 73 W (Type 4) usable power.

802.3af allows powered devices to draw up to about 13 W. Power is supplied as 350mA@48V and limited to 15.4 W, but the voltage drop over the maximum 100 feet of cable results in usable power of around 13 W.

802.11ac (Wi-Fi 5) is a standard for operating wireless access points and its communication with wireless devices. It is not a PoE-specific standard.

Related Content

5.2.10 Power over Ethernet

resources\questions\pt_core1_220_1201_2_5_006.question.xml

Question 73

✔ Correct

A user is trying to connect a gaming console to a television. What type of connector should the user utilize?

- ☐ Lightning port
- ☐ USB 3.0
- ☒ HDMI ✔ Correct
- ☐ Adapter

Explanation

A high-definition multimedia interface (HDMI) is the most widely used video interface. HDMI supports both video and audio. The user would utilize this because it is ubiquitous in consumer electronics.

USB 3.0 has new versions of Type A, Type B, and Type B Micro connectors with additional signaling and wires.

An adapter cable has connectors for two different cable types at the end. An active adapter uses circuitry to convert the signal.

A Lightning port is found only on Apple's mobile devices. The Lightning connector is reversible. The user needs a suitable adapter cable to connect such a device as a Lightning-to-USB A or Lightning-to-USB C.

Related Content

-  2.1.1 Personal Computers
-  2.1.3 Peripheral Devices
-  2.1.10 HDMI and DisplayPort Video Cables
-  2.2.12 Video Cards

resources\questions\pt_core1_220_1201_3_2_005.question.xml

Question 74

✔ Correct

A user is trying to print visual graphics for an upcoming meeting, but the paper is coming out faded. What should they check first?

☒ Toner cartridge ✔ Correct

☐ Creased paper

☐ Orientation

☐ Paper weight

Explanation

For faded or faint prints, if a simple cause, such as the user choosing an option for low density (draft output), can be discounted, then this is most likely to indicate that the toner cartridge needs replacing.

If the paper is not feeding into the printer, check that the paper is not creased, damp, or dirty.

If the printer is feeding multiple sheets also check the paper size and weight.

For incorrect page orientation, the paper size and orientation must be set correctly for the print job or the finishing/binding will be aligned to the wrong edge.

Related Content

10.3.3 Print Quality Issues

resources\questions\pt_core1_220_1201_5_6_002.question.xml

Question 75

✔ Correct

When a user downloads a Microsoft Store application, where does the application run to prevent the app from making system-wide changes and prevent a faulty store app from "crashing" the entire OS?

- ☐ Test development
- ☐ Resource requirements
- ☒ Sandbox ✓ Correct
- ☐ Cross-platform virtualization

Explanation

A sandbox is an isolated environment where Microsoft Store apps run to prevent the app from making system-wide changes and prevent a faulty store app from "crashing" the entire OS.

Test development often uses virtual desktop platforms based on some sort of guest OS hypervisor.

Cross-platform utilization tests software applications under different OSs and resource constraints and is important to ensure the application is compatible across several different environments.

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Related Content

8.1.2 Uses for Virtualization

resources\questions\pt_core1_220_1201_4_1_008.question.xml

Question 76

✔ Correct

A network engineer completes a site survey in an office for wireless connectivity. Given that the signal needs to get through several walls, what wireless frequency band would best suit the engineers' needs?

- ☐ Near-field communication
- ☒ 2.4 GHz ✓ Correct
- ☐ Bluetooth
- ☐ 5 GHz

Explanation

The 2.4 GHz standard is better at propagating through solid surfaces, thus giving it the most extended signal range.

The 5 GHz standard is less effective at penetrating solid surfaces and, therefore, does not support the maximum ranges achieved with 2.4 GHz standards.

Bluetooth is a short-range, wireless radio-network transmission medium that normally connects two personal devices, such as a mobile phone and a wireless headset.

Near-field communication(NFC) is the standard for two-way radio communications over very short (around four inches) distances, facilitating contactless payment and similar technologies.

Related Content 5.4.2 Frequency Bands 5.4.4 IEEE 802.11b/g

resources\questions\pt_core1_220_1201_2_2_002.question.xml

Question 77

✕ Incorrect

Which application protocol underpins file and printer sharing on Windows networks and usually runs directly over the TCP/445 port?

☐ Network Basic Input/Output System

☒ File Transfer Protocol ✕ Incorrect

☐ HyperText Transfer Protocol

☐ Server Message Block ✓ Correct

Explanation

Server Message Block (SMB) is the application protocol underpinning file and printer sharing on Windows networks. SMB usually runs directly over TCP port 445.

Network Basic Input/Output System (NetBIOS) allows computers to address one another by name and establish sessions for other protocols, such as SMB. This uses UDP port 137 for name services and TCP port 139 for session services.

The File Transfer Protocol (FTP) allows clients to upload and download files from a network server. It uploads files to websites and utilizes TCP ports 20 and 21.

HyperText Transfer Protocol (HTTP) is usually used to serve HTML web pages or applications hosted on a web server and typically operates over TCP port 80.

Related Content

 6.3.6 Well-Known Ports

 7.1.1 File/Print Servers

 7.1.9 Remote Terminal Access Servers

 14.2.1 Remote Desktop Tools

resources\questions\pt_core1_220_1201_2_2_009.question.xml

Question 78

✔ Correct

What are the causes of the error message, 'Boot device not found' ? (Select two.)

- ☐ Disk thrashing
- ☐ Grinding noise
- ☒ Faulty disk ✔ Correct
- ☒ File corruption ✔ Correct

Explanation

A faulty drive or connection to the drive can cause the error message, 'Boot device not found.'

Corruption of the boot section can cause the error message, 'Boot device not found.' An external bootable device can access the drive through the OS to attempt to fix the data or simply recover the data.

Disk thrashing is a constant read/write activity on the drive. Low random access memory (RAM), faulty software, or malware can cause disk thrashing to occur.

A grinding noise indicates mechanical problems with a hard disk drive (HDD). If the device is not initializing, it can not cause grinding noises.

Related Content

-  4.2.5 Troubleshoot Boot Issues
-  4.2.6 Lab: Troubleshoot Boot Issues
-  4.2.7 Troubleshoot Boot Sector Issues
-  4.2.9 Troubleshoot Drive Availability
-  4.2.10 Lab: Troubleshoot Drive Availability
-  14.4.8 Troubleshoot Boot Issues

resources\questions\pt_core1_220_1201_5_2_003.question.xml

Question 79

✔ Correct

What measures the PSU to meet the combined power requirements of the PC's components?

- ☐ Heat sink
- ☒ Wattage rating ✓ Correct
- ☐ Redundant power supply
- ☐ Modular power supply

Explanation

The wattage rating measures the power supply unit's (PSU's) output capability. Depending on the model, CPUs can range from 17 W to over 100 W. The power output is not the same as the PSU's power from grid power.

A redundant power supply requires a compatible motherboard. This configuration is more common in server systems than on desktop PCs. Typically, each PSU plugs into a backplane on a server and is hot-swappable.

A modular power supply has power connector cables that are detachable from the unit.

A heat sink is a block of copper aluminum with fins. The fins expose a larger surface area to the air around the component to achieve a cooling effect by convection.

Related Content

3.1.3 Wattage Rating

resources\questions\pt_core1_220_1201_3_6_003.question.xml

Question 80

✔ Correct

Sandboxes can be used to analyze viruses, worms, and Trojans. Sandboxes can also provide an isolated environment for what other important function?

- ☐ Cross-platform virtualization
- ☒ Testing and development ✓ Correct
- ☐ Legacy software/OS
- ☐ Resource requirements

Explanation

Test development often uses virtual desktop platforms based on some guest OS hypervisor.

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Cross-platform utilization tests software applications under different OSs and resource constraints and is important to ensure the application is compatible across several different environments.

Legacy software applications and operating systems may not perform well if a user's system has been upgraded.

Related Content

8.1.2 Uses for Virtualization

resources\questions\pt_core1_220_1201_4_1_002.question.xml

Question 81

✔ Correct

A penetration tester must infiltrate a company's facility to capture all the network traffic from a particular set of PCs over a physical network cable. The tester can only access the intermediate distribution frame (IDF) closets into which all devices are patched. What tool can the tester utilize to accomplish their goal?

☒ Network tap ✔ Correct

☐ Cable tester

☐ Wi-Fi analyzer

☐ Toner probe

Explanation

A network tap intercepts the signals passing over a cable and sends them to a packet or protocol analyzer. Taps are either powered or unpowered. There are passive and active capture types.

A Wi-Fi analyzer records statistics for the access point (AP) that the client is currently associated with and detects any other access points in the vicinity.

A toner probe detects the signal and follows the cable over ceilings and through ducts, or identifies it from the rest of the bundle.

A basic cable tester is a pair of devices attached to each cable end. It tests a patch cord or wall ports and panel patch ports connected via patch cords to test the permanent link.

Related Content

5.3.9 Network Taps

resources\questions\pt_core1_220_1201_2_8_005.question.xml

Question 82

✕ Incorrect

A user notices that their laser printer is not working and is causing an error. When the technician looks at the printer, they notice that the user was using thick paper, causing a part of the printer to overheat. Which part of the printer has overheated?

☐ Developer roller

☒ Pickup roller ✕ Incorrect

☐ Transfer roller

☐ Fuser unit ✓ Correct

Explanation

The paper passes through the fuser assembly after passing through the transfer assembly. The fuser unit squeezes the paper between a hot roller and a pressure roller so that the toner melts onto the paper's surface.

The transfer roller applies a positive charge to the underside of the paper. This causes the toner on the drum to be attracted to the paper. This prevents the paper from sticking to the drum or curling as it enters the fuser unit.

The pickup roller turns against the paper stack, pushing the paper into a feed and separation roller assembly.

The developer roller is a magnetized roller located very close to the photosensitive drum.

Related Content

resources\questions\pt_core1_220_1201_3_8_009.question.xml

Question 83

✔ Correct

A large store chain is eager to reduce the cost of maintaining its servers but is unsure what the best method would be for its use case. The major requirements for the chosen solution are that the store chain needs to own its cloud infrastructure (without sharing with another organization) to maintain maximum privacy and security of the data. Which cloud computing option best meets the store chain's major requirements?

☒ Private cloud ✔ Correct

☐ Hybrid cloud

☐ Public cloud

☐ Community cloud

Explanation

The private cloud meets the store chain's two major requirements because it gives organizations greater control over the privacy and security of its data stored in the cloud.

The public cloud is incorrect because there are risks regarding performance and security. The public cloud is a shared resource with other organizations, and a cloud service provider (CSP) runs it.

A hybrid cloud is incorrect because some cloud functions can be done via a public cloud, creating potential vulnerabilities to privacy and security.

Community cloud is incorrect because it requires the store chain to share a hosted private or fully private cloud with other organizations.

Related Content

8.2.3 Common Cloud Deployment Models

resources\questions\pt_core1_220_1201_4_2_001.question.xml

Question 84

✔ Correct

What component can be used in a camera to hold digital image storage?

- ☐ Optical drive
- ☐ Flash drive
- ☒ Memory card ✔ Correct
- ☐ HDD

Explanation

Memory card is a form factor used in consumer digital imaging products, such as digital still and video cameras, and to expand smartphone and tablet storage.

Flash drives are flash memory boards with a USB connector and protective cover.

Optical drives rate according to their data transfer speed. An optical drive can perform recording/rewriting and markets with three speeds.

A hard disk drive (HDD) stores data on metal or glass platters coated with a magnetic substance. The top and bottom of each platter are accessed by its own read/write head, moved by an actuator mechanism.

Related Content

-  2.2.4 Motherboard Storage Connectors
-  3.2.3 Solid-State Drives
-  3.2.10 Removable Storage Drives
-  3.2.11 Optical Drives
-  4.2.13 Troubleshooting Problems with Storage Drives & RAID Arrays
-  8.2.7 Cloud File Storage
-  12.4.2 Storage
-  15.2.2 Windows Domains and Active Directory
-  21.1.4 Backup Media Requirements

resources\questions\pt_core1_220_1201_3_4_008.question.xml

Question 85

✕ Incorrect

What causes a mobile device battery to swell?

☒ Improper charging routine ✕ Incorrect

☐ Faulty charging circuit ✓ Correct

☐ Under-powered adapter

☐ Old battery

Explanation

A swollen battery indicates a problem with the battery's charging circuit, which is supposed to prevent overcharging.

An improper charging routine will reduce the usable life of a battery. A Li-ion battery should not be allowed to fully discharge regularly or be kept persistently at 100% charge.

As batteries age, they will not hold a proper charge. Symptoms can be that the device will not switch on or only remains on for a few seconds.

Charging ports can be damaged by improper insertion and removal of the connectors. Signs of this are no charging or intermittent charging.

Related Content

9.4.1 Power and Battery Issues

resources\questions\pt_core1_220_1201_5_4_002.question.xml

Question 86

✕ Incorrect

A projector shut down two hours into a presentation. What could be the problem?

- ☐ Burned-out bulb
- ☒ Connectors not secure ✕ Incorrect
- ☐ Bad HDMI cable
- ☐ Overheating ✓ Correct

Explanation

Intermittent projector shutdown is typically caused by overheating. After being on for over two hours, overheating is a reasonable assumption. Check if the projector's fan is working and that the vents are free from dust and not obstructed.

A burned-out bulb will not turn the projector off but will instead cease to display an image.

A bad high-definition multimedia interface (HDMI) cable can cause problems, but the fact that the projector was working for over two hours and then shut down does not point to a cable problem.

Connectors that are not securely plugged in can generally be diagnosed by a flashing screen, not turning off.

Related Content

-  3.4.5 CPU Features
-  4.2.1 Troubleshoot Power Issues
-  4.3.2 Overheating
-  4.3.4 Troubleshoot Performance Issues
-  14.4.11 Troubleshoot Performance Issues

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Question 87

✕ Incorrect

A technician works on a faulty unit and needs to remove and replace a power supply unit (PSU) without opening the case and without the server losing power. What is this configuration?

- ☒ Open-loop system ✕ Incorrect
- ☐ Redundant power supply ✓ Correct
- ☐ Liquid-base cooling system
- ☐ Modular power supply

Explanation

A redundant power supply requires a compatible motherboard. This configuration is more common in server systems than on desktop PCs. Typically, each PSU plugs into a backplane on a server and is hot-swappable.

A modular power supply has power connector cables that are detachable from the unit.

A liquid-based cooling system refers to a system of pumping water around the chassis. Water is a more effective coolant than air convection, and a good pump can run more quietly than numerous fans.

An open-loop system will usually need draining, cleaning, and refilling periodically. The fans and radiators need to be kept dust-free. The technician should drain the system prior to moving the PC to a different location.

Related Content

 3.1.4 Power Supply Connectors

 3.1.7 Redundant Power Supplies

resources\questions\pt_core1_220_1201_3_6_001.question.xml

Question 88

✔ Correct

A user who is new to computers inspects a laptop to become acquainted with all the physical features. The user finds a small hole in the top bezel of the display. What feature is this?

- ☐ Webcam
- ☒ Microphone ✓ Correct
- ☐ LED Light
- ☐ Speaker

Explanation

On a laptop, the microphone is exposed by a small hole in the top bezel of the laptop display. This can be true with a laptop that either has or does not have a webcam built-in.

A digital camera/webcam allows for video recording or web conferencing and can also be used to take still pictures. The webcam is usually located on the top bezel of the display.

A light-emitting diode (LED) on a laptop is present to illuminate the subject during webcam use.

Speakers are usually found in either the side bezels of a laptop display or somewhere on the main laptop body.

Related Content

2.1.1 Personal Computers

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Question 89

✔ Correct

An IPv6 address is composed of which two main parts? (Select two.)

- ☐ Dynamic address
- ☒ Network ID ✔ Correct
- ☐ Static address
- ☒ Interface ID ✔ Correct

Explanation

An IPv6 address is in two main parts. The first 64 bits are a network ID.

An IPv6 address is in two main parts. The second 64 bits designate a specific interface.

Using static addressing requires an administrator to visit each computer to enter the configuration information for that host manually. This is traditionally only used for IPv4 addresses.

A host can receive its IP address, subnet mask, default gateway, and DNS server addresses from a Dynamic Host Configuration Protocol (DHCP) server as an alternative to static configuration. This is also used traditionally with only IPv4 addresses.

Related Content

6.2.15 IPv6 Addressing

resources\questions\pt_core1_220_1201_2_6_002.question.xml

Question 90

✔ Correct

How can a technician eliminate overheating as the cause of a sluggish computer?

- ☐ By touching the computer
- ☐ By running a speed test
- ☐ By checking for misconfigurations
- ☒ By checking the temperature sensor values to ensure they are normal ✓ Correct

Explanation

If the temperature is too high, the central processing unit (CPU) and other components are likely to throttle performance to avoid overheating. The technician can verify this by checking temperature sensor temps and fan speeds.

Even if the computer is not hot to the touch, it can still slow the computer down. Just touching the case to check the temperature of the CPU would not rule out temperature problems.

A slow computer can sometimes mean a network connection problem, which the technician can verify by running a speed test. This test would be appropriate after checking for temperature problems.

Misconfigurations can cause sluggish performance, but this would not help rule out overheating.

Related Content

-  3.4.5 CPU Features
-  4.2.1 Troubleshoot Power Issues
-  4.3.2 Overheating
-  4.3.4 Troubleshoot Performance Issues
-  14.4.11 Troubleshoot Performance Issues

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